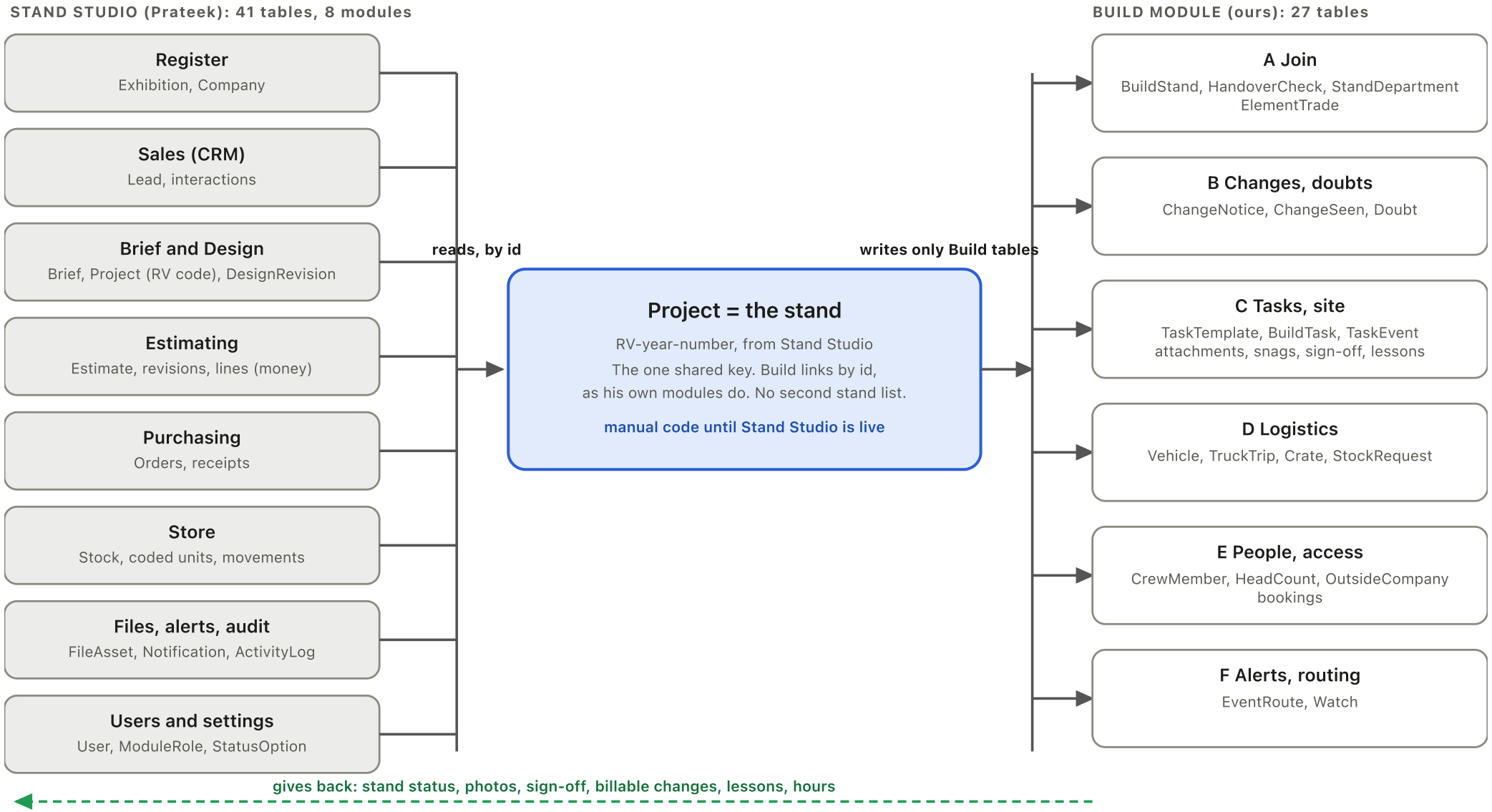
Two systems, one stand

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Prateek built Stand Studio for the Agency: lead, brief, design, estimate, purchasing, store. We add a Build module beside it. The Project, with its RV code, is the stand in both. Build reads his tables by id and writes only its own.



Our 27 tables (1 of 2)

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Groups A to C. Fields are in plain words with their pick-lists; a pick-list is a fixed set of values.

A. Join with Stand Studio

Table	What it is	Fields	Links by id to	Written by
BuildStand	One row per stand. The key is the Stand Studio Project id (by id).	projectId (unique), source (STAND_STUDIO or MANUAL), manualCode (only if MANUAL), buildStatus (pick), pmUserId, releasedAt, releasedById, goAheadAt, goAheadById	Project, User	PM, Aman
HandoverCheck	One row per stand per pack item: is it there?	standId, kind (TECHNICALS, COLOUR_SPEC, PRINT_PROOF, FINISH_SPEC, APPROVALS, PURCHASING_LIST), state (MISSING, PRESENT, WAIVED), sourceType, sourceId, waivedReason, checkedAt, checkedById	DesignRevision, FileAsset	PM; filled by the system when Agency records arrive
StandDepartment	Who has to see this stand: one row per stand per department.	standId, department (pick), reason (ELEMENT_RULE or MANUAL), neededFrom, neededBy, foremanUserId	User	System at release, PM edits
ElementTrade	Rule: which department a stand element needs (a lighting element needs Electric).	elementCode (an EstimatingCode value), department, note	EstimatingCode	Aman, Narendra

B. Changes and doubts

Table	What it is	Fields	Links by id to	Written by
ChangeNotice	A change that others must see. Insert-only.	standId, kind (DESIGN_REVISION, CLIENT_CHANGE, SPEC_CHANGE, DATE_CHANGE), designRevisionId, fromNo, toNo, summary, billable (yes/no, no price), severity (MINOR, MAJOR), raisedById, raisedAt	DesignRevision	System from Design uploads; the rep for client changes
ChangeSeen	One tap: a department has seen a change. Insert-only.	noticeId, department, userId, seenAt, onBehalfOfCrew (yes/no)	User	Foremen, PMs
Doubt	A question on a drawing and its answer.	standId, department, raisedById, designRevisionId, fileAssetId, markX, markY, voiceFileId, text (optional), status (OPEN, ANSWERED, CLOSED), answer, answeredById, answeredAt, changeNoticeId	DesignRevision, FileAsset, User	Trades raise, Design answers

C. Tasks and site

Table	What it is	Fields	Links by id to	Written by
TaskTemplate	The standard steps for each department and stage. Editable list.	department, stage (BUILD_AREA, SITE, CLOSE), code, label, order, appliesWhen (rule), usualHours (learned later)	none	Aman, heads
BuildTask	One step on one stand.	standId, templateId, partLabel, status (NOT_STARTED, WORKING, DONE, BLOCKED), blockReason (pick), startedAt, doneAt, doneById	TaskTemplate	Foremen, PMs (one tap)
TaskEvent	Every change to a task. Insert-only.	taskId, kind (START, DONE, BLOCK, UNBLOCK), byId, at, onBehalfOfCrew	User	System
BuildAttachment	A photo or voice note tied to a step.	fileAssetId, standId, ownerType (TASK, DOUBT, SNAG, CRATE, TEST, SIGNOFF), ownerId, takenAt, takenById	FileAsset (Stand Studio)	Whoever does the step
Snag	A defect found at handover.	standId, department, kind (pick), text (optional), raisedById, status (OPEN, FIXED, CLOSED), closedAt, closedById	User	PM, trades
Signoff	Client's sign-off at handover.	standId, signedAt, clientPersonName, method (ON_SCREEN, PAPER_PHOTO), fileAssetId, recordedById	FileAsset, User	Rep or PM
Lesson	Why something went wrong or well, in three taps.	standId, department, cause (pick), note (optional), recordedById, recordedAt	User	PM at close

Our 27 tables (2 of 2)

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Groups D to F, and what we reuse from Stand Studio instead of building.

D. Logistics and crates

Table	What it is	Fields	Links by id to	Written by
Vehicle	A truck or van.	code, kind, ownership (OWN or HIRED), plate, outsideCompanyId, active	OutsideCompany	Transport head
TruckTrip	One trip to a venue.	vehicleId, driverCrewId, exhibitionId, plannedAt, departedAt, reachedAt, offloadedAt, returnedAt	Exhibition	Transport
TripStand	Which stands ride on a trip.	tripId, standId	none	Transport
Crate	One labelled crate with a QR code.	standId, code (CR-year-4 digits, never reused), contents, packedByDepartment, packedAt, tripId, loadedAt, offloadedAt, returnedAt (each with the person)	none	Trades pack, Transport scans
StockRequest	Build asks Stand Studio's Store to move something. Build never writes the Store's own movement table.	standId, stockItemId, stockUnitId, kind (OUT or BACK), requestedById, state (REQUESTED, DONE)	StockItem, StockUnit	Transport, Electric, PM

E. People and access

Table	What it is	Fields	Links by id to	Written by
CrewMember	A person on the floor. A login is optional.	name, department, roleInTeam (FOREMAN, SENIOR, WORKER, DRIVER, LOADER, PM, COORDINATOR), employment (OWN or OUTSIDE), outsideCompanyId, userId (only those who sign in), phone (optional), active	User	Foremen, HR for the names
UserDepartment	Which department a login belongs to, and whether they head it.	userId, department, isHead	User	Aman
OutsideCompany	A company we hire from.	name, category (LABOUR, AV_LIGHTING, TRANSPORT, PM, PRINT, OTHER), contact, status (ACTIVE, BLOCKED), blockedReason	Supplier? (decision)	Aman
HeadCount	People on a stand on a day.	standId, onDate, department, ownCount, outsideCount, outsideCompanyId, enteredById (the latest row wins)	OutsideCompany	Foremen, Production Coordination
Resource	A machine or area that can be booked (CNC router, spray area).	code, label, section (from the machine brochure), active	none	Aman
Booking	A stand using a resource for a time.	resourceCode, standId, fromAt, toAt, bookedById	none	Foremen, Aman

F. Alerts and routing

Table	What it is	Fields	Links by id to	Written by
EventRoute	Who sees which event, and how. Editable by management without code.	eventKind, consumer (a department or a role), delivery (INSTANT, DIGEST, NONE), mustAck (yes/no), onlyIfAssigned (yes/no), digestHour	none	Narendra, Aman, top managers
Watch	A person who chooses to follow a stand, a department or all shows.	userId, scope (STAND, DEPARTMENT, ALL), standId, kinds, delivery	User	The person

Reused from Stand Studio

His table	How Build uses it
User, ModuleRole	Logins and roles. Build roles are added with module BUILD.
FileAsset	All uploaded files, photos and voice notes. Build links to it through BuildAttachment.
Notification	In-app alerts. Build adds new kinds only.

What we read, what we ask for, what we give back

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Three lists that make the join safe. Prateek's schema has changed seven times, so we read it through one thin layer and a written list of the fields we depend on.

Build reads (by id, read only)

Table	What we need	Why
Project	id, code (RV-year-number), outcome, technicalStatus (DRAFTING, CHECKING, APPROVED, RE_APPROVAL_REQUIRED), technicalApprovedAt, designerId	The stand's key and the gate: Build can release only when outcome is CONFIRMED and the technical is APPROVED. RE_APPROVAL_REQUIRED is the change signal.
Brief, Exhibition	clientName, venue, city, country, standNumber, standSize, clientDeadline; show name, eventFrom, eventTo, venue, hall, organiser	Everything about where and how big, so no one retypes it.
DesignRevision + FileAsset	revisionNumber, notes (what changed), uploadedBy, files (technical and per revision)	The design set and its versions. Build adds the Seen tap on top.
EstimateLine of the issued revision	elementCode, description, unit, quantity. Never unitCost, price or rates	The item list (the 'BOQ'). Build must see it without money.
PurchaseOrderLine, GoodsReceiptLine	projectId, description, quantity, neededBy, deliverTo (STORE or WORKSHOP), accepted	What is bought for the stand and what has arrived. Arrival tells the trade 'your material is here' with no message.
StockMovement, StockUnit, StockItem	OUT and BACK by projectId; RA-coded units	What stock and furniture went to the stand and came back.

We ask Prateek to add

#	Request	Why
1	A Coordination model: show windows (build-up, show, dismantle), loading-bay slots, passes, organiser and venue approvals. Stand Studio has none. Exhibition has only eventFrom and eventTo.	Build needs these dates by stand. Today they come by WhatsApp.
2	A price-free reading of the estimate lines (a view or an API that leaves out all money fields).	So Build gets the item list without prices.
3	The rep's go-ahead as a recorded action on the Project (goAheadAt, goAheadById).	The release gate needs it. The chart has it; the schema does not.
4	Events for technicalStatus changes and for each DesignRevision upload, written to ActivityLog (entity:action).	They are what Build's alerts are built from.
5	An element-to-trade mapping on EstimatingCode (or let Build keep it).	So a change to a lighting element alerts Electric and not everyone.
6	Permission for Build to attach files to FileAsset (a link table, or new link columns).	Photos from the floor must live in the same store.
7	A way to sign in for people without email, or agreement that foremen sign in and crews do not.	User needs an email and a password. Most workers have neither (assumed).
8	A manual stand: the right to create a Project-less BuildStand with a typed code until Stand Studio is live.	Our plan says Build must not wait for Stand Studio.

Build gives back

Data	Used by
BuildStand.buildStatus	The CRM and the rep's status page show where each stand is.
Photos, snags, sign-off	The rep's client status page and the final handover.
ChangeNotice with billable = yes	Estimating can reprice a client change; Finance sums it into the final invoice. Build holds no price.
Lessons	Design, through the PM.
Head count and hours (later)	Estimation can learn real hours per element. Hours, not money.

Who sees what, and when

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Every change writes an event; a routing list decides who is told. Nobody is told about what they do not touch. Management can edit the list without code.

Event	Comes from	Who sees it	How	Must tap Seen
project:confirmed	Agency (rep)	Aman, the PM if assigned	Instant	No
technical:approved	Agency (Design)	PM	Instant	No
technical:re_approval_required	Agency (Design)	PM, plus assigned departments as a warning	Instant	No
design:revision_uploaded	Agency (Design)	Only departments whose elements changed	Instant	Yes: Seen
coordination:window_changed	Agency (Coordination)	PM, then Transport through the PM	Instant	No
po_line:received	Agency (Procurement)	The department that needs the item; the PM	Digest	No
build:released	Build (PM)	Every assigned department, Transport (heads-up), Aman	Instant	Yes: Seen
build:doubt_raised	Build (trade)	Design to answer; PM copied	Instant to Design; digest to PM	No
build:doubt_answered	Design	The person who asked; the PM; everyone if it became a change	Instant	Yes if it became a change
build:task_done	Build (trade)	The next department waiting on it; the PM board	Instant to the next trade	No
build:task_blocked	Build (trade)	PM and the department head	Instant	No
build:step_late	System	PM first, then Aman after twice the usual time	Instant	No
build:trip_moved	Build (Transport scan)	PM; the rep's status page	Digest, instant on reached	No
build:signoff	Build or rep	Rep, Finance (for the invoice)	Instant	No
build:lesson	Build (PM)	Design, weekly	Digest	No
build:headcount_missing	System	The foreman, once at 10:00	Instant	No

Seen means the PM can see who has not tapped it yet. **Digest** means one short summary a day, so routine news does not interrupt work.

How it stays light for people

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Nine rules, and a time budget for each role. The budget is a target to test in a pilot, not a promise.

Nine rules

1. Enter once, at the source: forms start pre-filled from Stand Studio.
2. Arrival is automatic: a recorded delivery tells the trade its material is here.
3. Relevance, not broadcast: a change goes only to departments whose elements changed.
4. Quiet by default: routine news in one daily digest; instant only if it changes someone's current work.
5. One nudge, then escalate to the PM's board. No repeats.
6. Crews need no logins: foremen tap for them.
7. Works on site: taps and photos queue and send when there is a signal.
8. The PM is the screen, not the relay.
9. Link to Design's files; never copy them.

The burden budget

Role	Time and taps	What they do
Worker (no login)	0	The foreman taps for the crew. A worker is a name on a list, not a user.
Foreman	under 5 minutes a day (target, to verify)	Looks at My stands once. Taps Done on parts, adds a photo or two, one head count, taps Seen on a change. Never types a status.
Project manager	review the board 5 minutes; release check 10 minutes per stand, once	The board is built from others' taps. No weekly Excel, no WhatsApp to Sales and the boss. Site: one tap per step.
Aman	glance at the load board; one tap per allocation	Books outside labour in one short form. No gathering of PM lists.
Production Coordination (if they allocate daily work)	one head count screen	Enters who is on which stand once; everyone else reads it.
Transport	one scan per crate, three taps per trip	Loaded, reached and offloaded come from scans.
Design (Agency)	unchanged, plus answering doubts on the card	Uploads as today; the Seen tracking is automatic.
Rep (Agency)	go-ahead, client approvals, client changes	One button each; no forwarding.
Top managers	0	Read only.

What else we can digitize

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Ranked by what it gives against what it costs. Evidence says how we know it matters.

What	How	Value	Effort	Who gains	Evidence
Head count and hours, own and outside	QR or phone check-in per stand per day	High	Low	Planning, safety, cost; HR and Finance read it	ASSUMED: labour is recorded almost nowhere (told)
Machine and spray-area booking	Book CNC router, saws, spray area per stand	High	Low	Aman sees crowding before it happens. The machine brochure lists 45	ASSUMED
Crate and truck tracking	QR on crates; scans at load, arrival, offload, return	High	Medium	No 'where is it' calls; shows what is missing at site	ASSUMED
Equipment register for lights, cables, AV, tools	Reuse Store's RA-coded units; scan out and in	High	Medium	Reuse and loss; Electric and Procurement	ASSUMED
Print and cut queue (Printing, Acrylic, CNC)	File, size check, printed, packed	Medium-high	Low	No double printing	told: printed twice
Snags and on-screen sign-off with photos	Snag list by department; client signs on screen	High	Low	Faster handover; feeds the invoice	told: handover sheet on paper
Daily site report	Built from taps and photos, no writing	High	Low	Replaces the PM's messages upward	told: WhatsApp updates
Outside company register and booking	Who, what, when, status	Medium	Low	Aman; HR; Finance	told: Aman outsources
Planned versus actual hours per stand	From tasks and head counts	High later	Medium	Estimation learns real hours (hours, not money)	ASSUMED
Pipeline early warning	Quotes sent become likely stands by week	High	Low	Aman plans load before WON	ASSUMED
Show calendar and build-up windows	Coordination's dates, once	High	Medium	Agency owns it; Build depends on it	told: dates by WhatsApp
Safety and worker permits	Method statements, venue passes, labour permits	Medium-high	Medium	Venues may require them; Coordination and HR	ASSUMED; ask HR and Coordination
Damage and return condition	Condition grade on return (Store has this)	Medium	Low	Reuse decisions	told: most is scrapped
Scrap versus keep, with disposal record	One tap per item at dismantle	Medium	Low	Scrap cost visible; reuse tracked	told: dismantle and scrap
Lessons loop	Three-tap reasons at close, to Design	High	Low	Prateek: the missing loop	Prateek
Skills and machine permissions	Who may run CNC or spray	Low-medium	Low	Safety; HR	ASSUMED
Client status page	What the rep tells the client, from taps	Medium	Low	Replaces messages	ASSUMED
Vehicle log, delivery notes	Trips, fuel, maintenance, driver	Medium	Medium	Transport	ASSUMED

What does not fit yet, and what to decide

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Found while reading Stand Studio against what we know. Each needs a person to settle it.

Topic	What we see	Ask
BOQ	His EstimateLine is described as 'a line of the bill of quantities', yet his clash note says there is no per-stand BOQ. Both can hold: the BOQ exists in Estimating with money, and Build needs only the price-free item list. Name it once.	Prateek 3
Stand and Project	One Brief per Lead and one Project per Brief. Can one client have two stands at one show? If so, is that two leads?	Prateek 3
Doubts and DesignMessage	DesignMessage is a thread between sales and design. Build's Doubt is a different thread (trade to Design). Keep both; Design answers doubts on the Doubt card, and the rep can open it by link.	Prateek, Aman
Outside companies and Supplier	Supplier already has a category. Are outside labour and AV companies Suppliers, or a separate list? One owner only.	Prateek
Users with no email	User requires email and password. Foremen can sign in; most workers cannot. We use CrewMember for people without a login.	Prateek, Aman
Photos and storage	FileAsset keeps files under one upload folder. Photos from many phones add up fast. Who sizes the storage?	Prateek
Money	Estimating holds money fields marked money-class. Build roles must never read them.	Prateek
A moving schema	Seven updates so far. We read it through one thin layer and a written list of the fields we depend on, checked at every change.	Narendra, Prateek

Order of work (proposal)

1. **Slice 1: one current design, and who has seen it.** BuildStand, StandDepartment, ElementTrade, ChangeNotice, ChangeSeen, Doubt, EventRoute. Manual stands first; join to Stand Studio later.
2. **Slice 2: tasks, photos, boards.** BuildTask, TaskTemplate, TaskEvent, BuildAttachment, HandoverCheck, Watch.
3. **Slice 3: Transport.** Vehicle, TruckTrip, TripStand, Crate, StockRequest.
4. **Slice 4: people and capacity.** CrewMember, HeadCount, OutsideCompany, Resource, Booking.
5. **Slice 5: close-out.** Snag, Signoff, Lesson.

Honest estimate: no build time can be given yet. Slice 1 is a handful of small tables; the real work is agreeing the joins with Prateek and winning the habit.